3 **(a)** A student produces a transverse wave with a rope. The wave travels along the rope. Figure 3.1 shows the wave as seen from above.

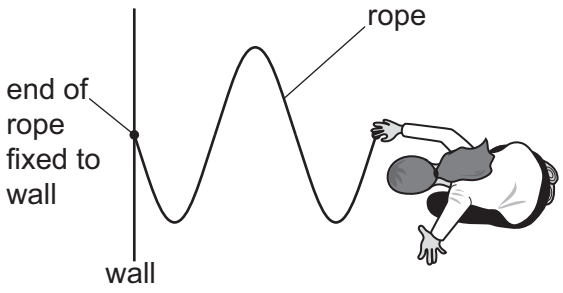


Figure 3.1

(i) Describe how the student produces the transverse wave.

.....

.....

..... [2]

(ii) The wavelength of the wave is 0.80 m.

Determine the distance between the wall and the student's hand.

distance = ... m [2]

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(b) A laser produces a beam of red light with a wavelength of 670 nm in air.

(i) State the approximate speed of light in air.

speed = ... m/s [1]

(ii) Calculate the frequency of the red light. State the unit.

frequency = ... unit ..... [3]

(iii) Complete the sentence.

Light of a single frequency is described as ..... light. [1]

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